Assessing Debt Sustainability

Liberia's external debt is very large and clearly unsustainable. Preliminary estimates based on incomplete data indicate that Liberia's external debt amounted to about US$3.7 billion in nominal terms at end-2005 (see Table 5 of the main text).[1] Of this total, it is estimated that US$1.6 billion were owed to multilateral creditors (of which US$740 million to the Fund), US$1 billion to the bilateral creditors (of which US$830 million to Paris Club members), and about US$1.1 billion (including accrued interest and penalty interest) to commercial creditors and suppliers. Almost all of this debt has been in prolonged arrears.

Calculations using the low-income country external debt template indicate that, at end-2005, the NPV of Liberia's external debt relative to GDP and exports of goods and services was 790 percent and 3,040 percent, respectively (Tables A1 and A2, Figure A1). These ratios are many times higher than the relevant debt burden thresholds of the LIC DSA framework.[2]

Under a baseline scenario that represents an extension of the staff's medium-term baseline scenario, the debt burden indicators decline over time owing to the assumed strong growth of the economy and exports but never come close to the debt burden thresholds.[3] Thus, the NPV of debt-to-GDP ratio declines to 460 percent by 2010 and 240 percent by 2024, while the NPV of debt-to-exports ratio declines to 1,540 percent by 2010 and 780 percent by 2024.

Alternative scenarios and stress tests underscore the unsustainability of Liberia's external debt.[4] Under a historical scenario, which sets important variables at their average values for 1999–2004, the debt burden indicators decline at a slower pace than in the

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[1] Most of Liberia's official records on external debt were lost during the years of conflict and the process of obtaining statements from creditors has not been completed. Data on debt to multilateral creditors are relatively complete but less so for bilateral and particularly commercial creditors. The latter reflect estimates by the authorities and staff.

[2] The thresholds for a country ranked as a poor policy performer according to the World Bank's CPIA index, as Liberia is, are 30 percent and 100 percent, respectively

[3] Under this baseline scenario, real GDP is assumed to grow at about 8 percent per year during 2006-10 and 5 percent per year thereafter. Exports are assumed to expand rapidly during 2006-10 as timber exports recover following the lifting of UN sanctions. Moreover, Liberia is assumed to continue to accumulate arrears and not have access to any new borrowing for the entire projection period.

[4] At this stage, it would be premature to present a scenario incorporating possible debt relief because of uncertainty regarding the timing of arrears clearance. This will depend on the mobilization of adequate financing assurances, and Liberia's success in building a track record of macroeconomic management and structural reforms. However, if and when debt relief becomes available, Liberia's debt sustainability outlook would, of course, improve dramatically.

baseline. Thus, the NPV of debt-to-GDP and to-export ratios are at 470 percent and 1,560 percent, respectively, in 2024, about twice their levels under the baseline. Under the most extreme stress test, represented by a one standard deviation shock to exports in 2006–07, the debt burden indicators first rise sharply and then converge slowly to the levels under the historical scenario, despite the larger decline in the stock and flow indicators relative to the baseline projections.

The application of the LIC DSA framework to the case of Liberia clearly illustrates that country's external debt is unsustainable situation. To address its external debt problem, Liberia will need to demonstrate a strong commitment to sound macroeconomic and structural policies but will also require substantial support, including exceptionally deep debt relief, from the international community.

Table A1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2002-2024 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Estimate					Projections				
	2002	2003	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2014	2024	2011-24 Average
External debt (nominal) 1/	**545.5**	**772.8**	**751.9**			**667.3**	**605.4**	**558.2**	**512.1**	**469.4**	**426.2**		**353.4**	**236.0**	
o/w public and publicly guaranteed (PPG)	545.5	772.8	751.9			667.3	605.4	558.2	512.1	469.4	426.2		353.4	236.0	
Change in external debt	-0.3	227.2	-20.9			-84.6	-61.9	-47.2	-46.1	-42.7	-43.2		-16.1	-8.8	
Identified net debt-creating flows	-13.8	174.9	-80.6			-35.1	-30.9	-30.0	-29.2	-27.2	-29.6		-6.9	-1.1	
Non-interest current account deficit	**-6.6**	**7.2**	**2.2**	**6.2**	**10.1**	**-13.4**	**-5.5**	**2.2**	**1.2**	**0.3**	**0.8**	-2.4	**5.6**	**6.4**	6.0
Deficit in balance of goods and services	-3.8	7.1	33.1			29.4	31.4	30.4	27.8	24.2	19.2		19.2	19.2	
Exports	37.3	31.9	27.4			25.9	22.9	22.3	23.9	26.3	30.1		30.1	30.1	
Imports	33.5	39.1	60.5			55.4	54.3	52.7	51.7	50.5	49.4		49.4	49.4	
Net current transfers (negative = inflow)	-11.3	-10.7	-48.8	-18.4	15.4	-56.7	-51.7	-45.7	-41.4	-37.6	-31.2	-44.1	-27.3	-19.6	-24.5
Other current account flows (negative = net inflow)	8.5	10.7	17.9			14.0	14.8	17.6	14.8	13.7	12.8		13.6	6.8	
Net FDI (negative = inflow)	**-1.3**	**1.4**	**1.8**	**-1.9**	**3.2**	**-1.1**	**3.7**	**-4.4**	**-3.6**	**-2.5**	**-2.6**	-1.7	**-2.6**	**-2.6**	-2.6
Endogenous debt dynamics 2/	**-5.9**	**166.4**	**-84.6**			**-20.6**	**-29.1**	**-27.7**	**-26.8**	**-25.1**	**-27.8**		**-9.9**	**-5.0**	
Contribution from nominal interest rate	10.0	11.0	11.0			15.2	16.3	15.3	14.2	13.1	12.1		7.3	6.5	
Contribution from real GDP growth	-19.5	219.4	-17.3			-35.9	-45.3	-43.0	-41.0	-38.2	-39.9		-17.3	-11.4	
Contribution from price and exchange rate changes	3.6	-64.0	-78.3			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**13.5**	**52.3**	**59.7**			**-49.5**	**-31.0**	**-17.2**	**-16.9**	**-15.5**	**-13.6**		**-9.1**	**-7.7**	
o/w exceptional financing	-15.0	-16.7	-16.9			-18.3	-18.9	-17.7	-16.3	-14.6	-12.7		-9.7	-4.9	
NPV of external debt 4/	...	932.2	900.0			788.6	702.1	632.6	568.8	520.1	463.5		367.9	236.5	
In percent of exports	...	2919.4	3282.3			3040.6	3068.5	2832.9	2381.6	1977.5	1537.4		1220.3	784.3	
NPV of PPG external debt	**...**	**932.2**	**900.0**			**788.6**	**702.1**	**632.6**	**568.8**	**520.1**	**463.5**		**367.9**	**236.5**	
In percent of exports	...	2919.4	3282.3			3040.6	3068.5	2832.9	2381.6	1977.5	1537.4		1220.3	784.3	
Debt service-to-exports ratio (in percent)	36.0	48.6	54.8			71.9	84.3	80.6	70.1	56.3	43.5		26.9	23.8	
PPG debt service-to-exports ratio (in percent)	36.0	48.6	54.8			71.9	84.3	80.6	70.1	56.3	43.5		26.9	23.8	
Total gross financing need (billions of U.S. dollars)	0.0	0.1	0.1			0.0	0.1	0.1	0.1	0.1	0.1		0.1	0.3	
Non-interest current account deficit that stabilizes debt ratio	-6.3	-220.1	23.1			71.2	56.4	49.4	47.3	43.1	44.0		21.6	15.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.7	-31.3	2.6	0.7	20.4	5.3	7.7	8.0	8.3	8.4	9.6	7.9	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	-0.7	13.3	11.3	3.8	8.2	4.9	5.7	3.8	4.0	3.9	3.6	4.3	1.9	1.9	1.9
Effective interest rate (percent) 5/	1.9	1.6	1.6	2.5	1.1	2.2	2.8	2.8	2.9	2.9	2.9	2.8	2.1	2.8	2.3
Growth of exports of G&S (US dollar terms, in percent)	29.3	-33.3	-2.0	13.4	39.5	4.4	0.4	9.4	20.5	24.0	30.2	14.8	7.0	7.0	7.0
Growth of imports of G&S (US dollar terms, in percent)	-0.5	-9.2	76.6	10.7	37.5	1.1	11.6	8.9	10.5	10.0	10.9	8.8	7.0	7.0	7.0
Grant element of new public sector borrowing (in percent)	...	...	10.9	...	...	10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9
Memorandym item:															
Nominal GDP (billions of US dollars)	0.6	0.4	0.5			0.5	0.6	0.7	0.8	0.9	1.0		1.3	2.6	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

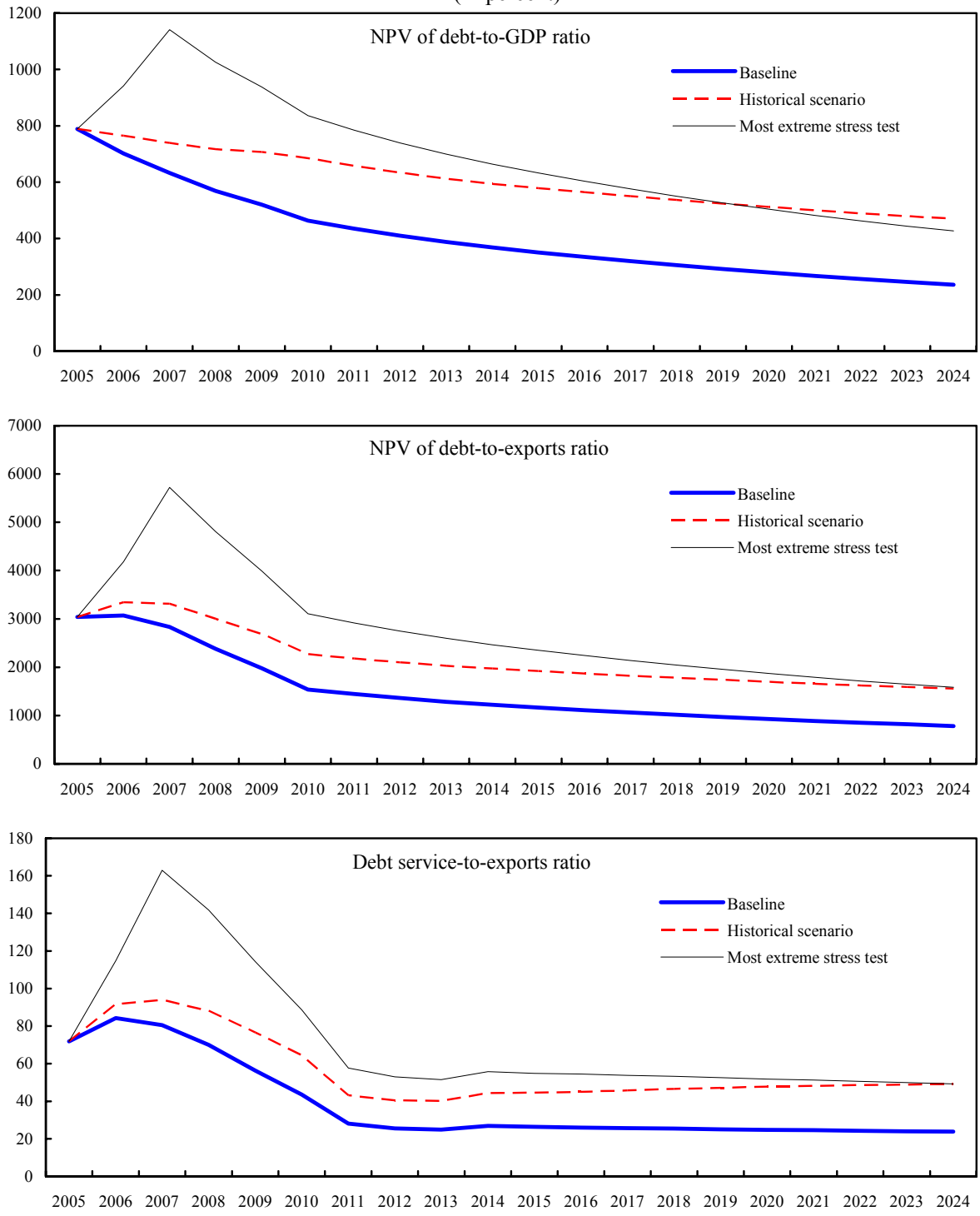
Table A2. Liberia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-24
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	**2014**	2024
NPV of debt-to-GDP ratio								
Baseline	789	702	633	569	520	464	**368**	236
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-24 1/	789	767	743	723	714	694	**607**	492
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	789	942	1141	1026	938	836	**664**	427
B2. Export value growth at historical average minus one standard deviation in 2006-07 2/	789	704	635	571	522	465	**369**	237
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	789	776	759	682	624	556	**441**	284
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 3/	789	720	668	601	550	491	**392**	249
B5. Combination of B1-B4 using one-half standard deviation shocks	789	908	1040	936	856	764	**610**	388
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 4/	789	980	883	794	726	647	**514**	330
NPV of debt-to-exports ratio								
Baseline	3041	3069	2833	2382	1978	1537	**1220**	784
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-24 1/	3041	3352	3329	3025	2715	2301	**2013**	1632
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	3041	3069	2833	2382	1978	1537	**1220**	784
B2. Export value growth at historical average minus one standard deviation in 2006-07 2/	3041	4185	5720	4809	3993	3105	**2466**	1583
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	3041	3069	2833	2382	1978	1537	**1220**	784
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 3/	3041	3147	2991	2518	2091	1628	**1301**	826
B5. Combination of B1-B4 using one-half standard deviation shocks	3041	3372	3721	3132	2602	2025	**1617**	1028
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 4/	3041	3069	2833	2382	1978	1537	**1220**	784
Debt service ratio								
Baseline	72	84	81	70	56	44	**27**	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-24 1/	72	92	95	89	77	65	**45**	52
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	72	84	81	70	57	44	**28**	24
B2. Export value growth at historical average minus one standard deviation in 2006-07 2/	72	115	163	142	114	89	**56**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	72	84	81	70	57	44	**28**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 3/	72	84	82	73	59	46	**29**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	72	91	103	91	73	57	**36**	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 4/	72	84	81	70	57	44	**28**	24
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 5/	61	61	61	61	61	61	**61**	61

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
3/ Includes official and private transfers and FDI.
4/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
5/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure A1. Liberia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2004-2024
(In percent)



Source: Staff projections and simulations.